Khosrow Ebrahimi
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EDUCATION

Pos*tdoctoral Research Associate* **March 2012 – Present**
Villanova University, Villanova, PA

- Research Title: Thermal Management and Waste Heat Recovery in Data Centers
- Advisors: Dr. Amy S. Fleischer, Dr. Gerard F. Jones, and Dr. Aaron P. Wemhoff

Ph.D., Mechanical Engineering **May, 2012**
Kansas State University, Manhattan, KS

- **GPA: 3.9**
- Dissertation Title: Numerical simulation of turbulent airflow, tracer gas diffusion, and particle dispersion in a mockup aircraft cabin
- Advisors: Dr. Mohammad H. Hosni and Dr. Zhongquan C. Zheng

MS, Mechanical Engineering **August, 2003**
K.N. Toosi University of Technology, Tehran, Iran

- **GPA: 3.5**
- Thesis Title: A numerical study of forced convective heat transfer to turbulent pipe flow of supercritical water
- Advisor: Dr. Majid Bazargan

BS, Mechanical Engineering **September, 2000**
University of Tehran, Tehran, Iran
BS Project Title: Study of design and manufacturing procedures for thermoplastic and bakelite moulds

PROFESSIONAL & EXECUTIVE EXPERIENCE

CFD Consultant **August 2014 – Present**
Mechanical Engineering Department, Villanova University, Villanova, PA

- Project Title: Computational thermal simulation of LED lamps Aug. 2014 - Dec. 2014
  - Client: Schiff Hardin LLP
- Project Title: CFD simulation and design optimization of orifice valves Sept.2014 – March 2015
  - Client: BorgWarner Inc.
- Project Title: CFD simulation of ridged tube configuration April 2015 - Present
  - Client: GLTT-US

Research Faculty **March 2012 – Present**
Center for Energy Smart Electronic Systems (ES2), Villanova University, Villanova, PA

- Primary researcher on an NSF/Industry sponsored research project for recovery and reuse of waste heat from large data centers
- Identify the optimal techniques to repurpose the low quality but high quantity energy dissipated by data centers
- In-depth thermo-economic analysis of novel heat recovery systems through the integration of absorption refrigeration and organic Rankine cycle into server cooling system

- Collaboration in developing thermodynamic tools for holistic analysis and optimization of energy efficient data centers
- Attend regular meetings with top engineers from the leading industrial data center operators including Comcast, IBM, Microsoft, and Facebook

Graduate Research Assistant **August 2008 - March 2012**
Mechanical and Nuclear Engineering Department, Kansas State University, Manhattan, KS

- 3D CFD simulation of turbulent airflow in an 11-row Boeing 767 aircraft cabin mockup
- 3D CFD simulation of turbulent airflow, tracer gas diffusion and particle dispersion in a generic half cabin model
- Numerical study of 2D and 3D backward facing step flow
- Extensive and professional application off FLUEN 6.3, FLUENT ANSYS and GAMBIT
- Collection and organization of the experimental data produced by the Aircraft Cabin Environmental Research (ACER) for the formal report

Department Head of Fuel Economy Standards in Transportation Sector **Jan. 2006 - May 2008**
Iranian Fuel Conservation Company (IFCO), Tehran, Iran

- Participation in developing a standard driving cycle for Tehran
- Review and evaluation of technical proposals submitted to IFCO containing technical solutions to improve the fuel efficiency in M1 and N1 light duty vehicles
- Lead projects to measure fuel consumption and emissions of retrofitted bi-fuel vehicles (run on gasoline and CNG) of Iran's light duty vehicles, as part of a comprehensive feasibility study for setting fuel economy standards for this type of vehicles
- Responsible for managing a team of four specialists, including two Master's degree holders in Mechanical Engineering, one Master's degree holder in Chemical Engineering, and one Bachelor's degree holder in Environmental Engineering
- Senior member of the Iran's National Committee for Setting the Fuel Economy Standards in Transportation Sector
- Member of the Technical Approval Committee for verifying and issuing the Type Approval (TA) for the Imported Motor Vehicles according to fuel consumption/emission standards
- Revision and update of ISIRI 4241-2 (The national fuel consumption standard and energy labeling for the Light Duty Gasoline Vehicles)
- Finalization and issuance of the first edition of ISIRI 8361 (The national fuel consumption standard and energy labeling for the Heavy Duty Vehicle Diesel Engines)
- Participation in design and establishment of the Motorcycle emission/fuel consumption measurement lab in the Energy Research Center of the Research Institute of Petroleum Industry (RIPI) with the collaboration of CP Engineering (UK) and AVL (Austria)
- Supervision and participation in annual preparation and publication of the Iran's Fuel Consumption Guide (FCG)
- Participation in preparing the annual reports for the Energy Commission of the parliament of Iran
- Preparation official proposals for the Management and Planning Organization (MPO) of Iran (until 2007) and National Iranian Oil Company (NIOC) board of directors in order to dedicate financial resources to run research projects targeted in fossil fuel energy saving in transportation sector

Project Engineer **October 2003 - December 2005**
Iranian Fuel Conservation Organization (IFCO), Tehran, Iran

- Compilation of ISIRI 4241-2 (The national standard for fuel consumption of Light Duty Vehicles) and ISIRI 8361 (The national standard for fuel consumption of Heavy Duty Vehicle Diesel Engines)

- Preparation and publication of the first edition of Iran's Fuel Consumption Guide (FCG)
- Member of the Iran's National Committee for Setting the Fuel Economy Standards in Transportation Sector
- Representative of IFCO in Global Energy Council sessions in Iran
- Study and test of 323 in-use light duty M1 and N1 vehicles of Iran's light duty vehicles for fuel consumption and emission (based on ECE R 83.02 standard driving cycle)
- Study and test of 53 Iranian newly manufactured light duty M1 and N1 vehicles for fuel consumption and emission (based on ECE R 83.02 standard driving cycle)
- Study of SAE J826 (H-Point Manikin) and 40 CFR 600.315-82(classes of comparable automobiles), in order to use in classifying Iranian light duty vehicles for setting fuel economy standard purposes
- Study and test of the Iranian newly manufactured Heavy Duty Vehicles' diesel engines for fuel consumption and emission (based on ECE R 49 standard driving cycle)

Research Assistant **September 2000 - August 2003**
Faculty of Mechanical Engineering, K. N. Toosi University of Technology, Tehran, Iran

- Development of a mathematical model to simulate forced convection heat transfer to turbulent flow of supercritical water

Intern Engineer **Summers of 1998 & 1999**
SAIPA (Societe Annonyme Iranienne de Production Automobile) Automotive Industries, Tehran, Iran

- Study of metal formation methods in automotive industries with a special focus on roll-forging
- Study of design and manufacturing procedures for thermoplast and backelite moulds

TEACHING EXPERIENCE

Lab Instructor **September 2013 – January 2014**
Mechanical Engineering Department, Villanova University, Villanova, PA

- Thermo-Fluid Lab Instructor for solar flat plate collector and fuel cell experiments: Responsible for teaching the theory and practice of experiments, taking quizzes, and grading lab reports for ME undergrad students

Graduate Teaching Assistant **January 2010 - May 2011**
Kansas State University, Mechanical and Nuclear Engineering Department, Manhattan, KS

- Teaching Assistant for Heat transfer (ME573, 3 credit hours): Responsible for weekly help sessions, grading homeworks, quizzes and midterm exams for 40 undergraduate students enrolled every semester

Teaching Student **September 2000 - August 2003**
K.N. Toosi University of Technology, Faculty of Mechanical Engineering, Tehran, Iran

- Teaching Assistant for Fluid Mechanics (3 credit hours), Engineering Thermodynamics (3 credit hours), Heat Transfer (3 credit hours): Responsible for grading homeworks, quizzes, midterm and final exams for 40 undergraduate students enrolled every semester

PUBLICATIONS

Published journal articles:

[1] Ebrahimi, K., Jones, G.F., Fleischer, A.S.,2015, " Thermo-economic analysis of steady state waste heat recovery in data centers using absorption refrigeration", Applied Energy, Vol. 139 , pp. 384-397.

[2] Ebrahimi, K., Jones, G.F., Fleischer, A.S.,2014, " A review of data center cooling technology, operating conditions and the corresponding low-grade waste heat recovery opportunities", Journal of Renewable and sustainable Energy Reviews, Vol. 31, pp. 622-638.

[3] Ebrahimi, K., Zheng Z., and Hosni, M.H., 2013, " A computational study of turbulent airflow and tracer gas diffusion in a generic aircraft cabin model, ASME Journal of Fluids Engineering, Vol.135, pp. 111105:1-15.

Under-preparation and submitted journal articles:

[1] Ebrahimi, K., Zheng Z. and Hosni, M.H., " A computational study of turbulent dispersion of micron-sized spherical particles in a generic aircraft cabin model", International Journal of Multiphase Flow (under-preparation).

[2] Ebrahimi, K., Jones, G.F., Fleischer, A.S., " A comprehensive thermo-economic analysis of steady-state waste heat recovery in data centers using organic Rankine cycle", Journal of Energy (under-preparation).

[3] Wemhoff, A.P., Ebrahimi, K., Jones, G.F., Fleischer, A.S., " Integration of waste heat recovery modules to the chip to cooling tower thermodynamic system simulator (VTAS)", Journal of Applied Thermal Engineering (under-preparation).

Peer- reviewed conference papers:

[1] Ebrahimi, K., Jones, G.F., Fleischer, A.S., 2014, " A review of challenging issues in the integration of absorption refrigeration and organic Rankine cycle into the data center cooling system", The IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems (ITHERM 2014), Paper No. 156, May 27-30, 2014,Orlando, FL

[2] Ebrahimi, K., Hosni, M.H., and Zheng, Z. C., 2013, " Computational study of turbulent airflow in a full-scale aircraft cabin mockup," ASME 2013 Fluid Engineering Summer Meeting, Paper number FEDSM2013-16564, July 7-11, 2013, Incline Village, NV.

[3] Ebrahimi, K., Zheng, Z. C., Hosni, M.H., 2012, " Computational study of the effects of particle size and density, particle injection and pressure on turbulent dispersion of micon-sized particles in a generic aircraft cabin," ASME 2012 International Mechanical Engineering Congress & Exposition, Paper number IMECE2012-88504, Nov. 9-15, 2012, Houston, TX.

[4] Ebrahimi, K., Zheng, Z. C., and Hosni, M.H., 2011, " Simulation of Turbulent Dispersion of 10 Micron Particles in a Generic Half-Cabin Model," ASME 2011 International Mechanical Engineering Congress & Exposition, Paper number IMECE2011-64183, November 11-17, 2011, Denver, CO.

[5] Ebrahimi, K., Zheng, Z. C., and Hosni, M.H., 2010, " LES and RANS Simulation of Turbulent Airflow and Tracer Gas Injection in a Generic Aircraft Cabin Model," 2010 ASME Fluids Engineering Division Summer Meeting and Exhibition, Paper number FEDSM2010-30552, August 1-4, 2010, Montreal, Canada.

[6] Ebrahimi, K., "Experimental investigation of using Closed Loop Carburetor (CLC) on Pride* and PK* wheel power, fuel consumption and emission", Proceedings of the 1st International Conference on Energy Saving and Energy Efficiency in Transportation Sector, Tehran, Iran, 2004.

[7] Ebrahimi, K., "Experimental study of the effect of the age, regular technical inspection and maintenance on Paykan* fuel consumption and emission situation", Proceedings of the 1st International Conference on Energy Saving and Energy Efficiency in Transportation Sector, Tehran, Iran, 2004.

[8] Ebrahimi, K. and Bazargan, M., "Numerical investigation of radial variation of heat flux and shear stress on heat transfer to turbulent flow of supercritical water", Proceedings of ISME 2004 8th International Conference of Mechanical Engineering, Tehran, Iran, 2004.

[9] Ebrahimi, K. and Bazargan, M., "Analytical study of forced convection heat transfer to turbulent flow of supercritical water", Proceedings of the 8th National Conference on Fluid Dynamics, Tabriz, Iran, 2003.

Poster Presentations:

- <u>Ebrahimi, K.</u>, Fleischer, A.S., Jones, G.F., "A Comprehensive Assessment of Technologies Utilizing Waste Heat and their Impact on Improving Energy Efficiency in Data Centers", NSF-IUCRC/ IAB Meeting, Villanova, PA, USA, June 2012.
- <u>Ebrahimi, K</u>., Fleischer, A.S., Jones, G.F., "Approaches for Recovery of Waste Energy from Data Centers", IUCRC IAB Meeting, National Science Foundation (NSF), Arlington, TX, USA, October 2012.
- <u>Ebrahimi, K</u>., Fleischer, A.S., Jones, G.F., "Waste heat recovery and reuse from Data Centers", NSF-IUCRC/ IAB Meeting, Atlanta, GA, USA, April 2013.
- <u>Ebrahimi, K</u>., Bakhshi, R., Fleischer, A.S., Jones, G.F., "Waste energy recovery and reuse from Data Centers-Absorption Refrigeration Cycle", NSF- IUCRC/ IAB Meeting, Villanova, PA, USA, September 2013.
- <u>Ebrahimi, K</u>., Bakhshi, R., Fleischer, A.S., Jones, G.F., "Waste energy recovery and reuse from Data Centers-Organic Rankine Cycle", NSF-IUCRC/ IAB Meeting, Villanova, PA, USA, September 2013.

Selected Talks:

- ES2 IU/CRC IAB meeting, Villanova, PA (2013)
- ES2 IU/CRC IAB meeting, Arlington, TX (2012)
- IMECE conference, Denver, CO (2011)

* Domestic automobile models

EDITORIAL ACTIVITIES

- Reviewer for the International Journal of Electrical Power and Energy Systems 2014
- Reviewer for the ASME Journal of Electronic Packaging 2013
- Reviewer for ASME 2013 Fluids Engineering Division Summer Meeting 2013
- Reviewer for the journal of Science of the Total Environment 2012
- Reviewer for ASME 2012 International Mechanical Engineering Congress & Exposition 2012

SKILLS

- Programming: FORTRAN, MATLAB and C
- Professional Softwares: ANSYS FLUENT, Gambit, SolidWorks, Mathcad, Tecplot
- Application Softwares: Microsoft Word, Excel, Power Point, Latex

DISTINCTIONS

- Postdoctoral research was selected for inclusion in the 2014 Breakthroughs of NSF Industry/University Cooperative Research Centers provided to members of Congress, White House and congressional staffers.
- Winner of poster contest in ES2 IU/CRC meeting, Villanova, PA (Sept 2013)
- Employed as an outstanding alumnus by the National Iranian Oil Company (Oct 2003)
- Ranked 3rd among 40 graduate students who admitted in 2000 by the Faculty of Mechanical and Aerospace Engineering K.N. Toosi University of Technology (Aug 2003)
- Ranked 207th among more than 5000 BS holders who participated in Iranian nationwide graduate schools entrance exam for mechanical and aerospace engineering masters programs(Sept 2000)
- Ranked 523rd amongst more than one million high school graduates who participated in Iranian nationwide university entrance exam (Sept 1995)